

SMARTBRIDGE / SKILL WALLET

Performance Testing Report

Date	2 July 2026	Team ID	—
Project Name	CaptionAI: AI-Powered Social Media Caption Generator		

Performance testing is critical to ensure your solution is fast, scalable, and stable under specific workloads. Document your testing strategy, the tools utilized, and the specific metrics measured against your target thresholds.

Step 1: Test Scenarios & Tooling

S.No	Test Type	Scenario / API Endpoint Tested	Tool(s) Used
1	Load Testing	Simulating concurrent requests hits on the core asynchronous dashboard endpoints to verify baseline application stability under steady user interaction loads.	Locust, Postman
2	Stress Testing	Pushed the Caption Generator backend layer and Groq Cloud inference API pipelines past normal operational ceilings to evaluate breaking points.	Locust
3	Spike Testing	Observed the network gateway stability and regex payload parsing engines under a sudden burst of high-volume text analysis workloads.	Locust, Ngrok Agent
4	Endurance Testing	Ran regular background execution request loads on the Python Flask runtime for extended intervals to observe long-term stability and potential leaks.	Locust

Step 2: Key Performance Indicators (KPIs) & Results

S.No	Metric (KPI)	Target Threshold	Actual Result	Status
1	Average Response Time	< 2000 ms	1850 ms	Pass
2	Throughput	> 20.0 RPS	22.4 RPS	Pass

S.No	Metric (KPI)	Target Threshold	Actual Result	Status
3	CPU Utilization	< 50% under load	34%	Pass
4	Memory (RAM) Utilization	< 60% under load	48%	Pass
5	Error Rate	< 1%	0.0%	Pass